

Multiple Choice

1. **Answer:** C

Options: A) Best-subset selection; B) Forward stepwise selection; C) Forward stage wise selection; D) All of the above

Note: Correct option: C - Forward stage wise selection

2. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

3. **Answer:** D

Options: A) Increase bias; B) Decrease bias; C) Increase variance; D) Decrease variance

Note: Correct option: D - Decrease variance

4. **Answer:** B

Options: A) pure; B) not pure; C) useful; D) useless

Note: Correct option: B - not pure

5. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: D - False, True

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Lower variance'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'O(D)'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Step size is too large'.

Long Form Response

11. **Answer:** `def find_Min_Diff(arr,n): | return diff`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def odd_position(nums): | return all(nums[i]%2==i%2 for i in range(len(nums)))`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key